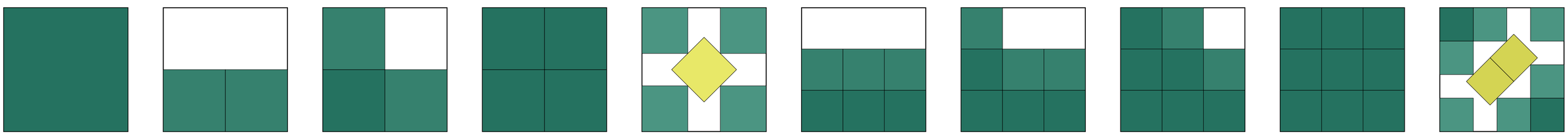
100 BEST KNOWN SQUARE PACKINGS

The Squares Project · github.com/jlevy/squares



1 O = R
s(1) = 1

2 O =
s(2) = 2

3 O =
s(3) = 2

4 O = R
s(4) = 2

5 O = R
s(5) = 2.707107 deg 2

6 O =
s(6) = 3

7 O =
s(7) = 3

8 O =
s(8) = 3

9 O = R
s(9) = 3

10 O =
s(10) = 3.707107 deg 2

11 O = R
s(11) = 3.877084 deg 8

12 =
s(12) ≤ 4
s(12) ≥ 3.971500

13 O =
s(13) = 4

14 O =
s(14) = 4

15 O =
s(15) = 4

16 O = R
s(16) = 4

17 =
s(17) ≤ 4.675531 deg 18
s(17) ≥ 4.660442

18 =
s(18) ≤ 4.822876 deg 2
s(18) ≥ 4.705000

19 =
s(19) ≤ 4.885619 deg 2
s(19) ≥ 4.825000

20 =
s(20) ≤ 5
s(20) ≥ 4.905000

21 O =
s(21) = 5

22 O =
s(22) = 5

23 O =
s(23) = 5

24 O =
s(24) = 5

25 O = R
s(25) = 5

26 =
s(26) ≤ 5.621321 deg 2
s(26) ≥ 5.545000

27 =
s(27) ≤ 5.707107 deg 2
s(27) ≥ 5.643500

28 = R
s(28) ≤ 5.824445 deg 6
s(28) ≥ 5.735000

29 ≈
s(29) ≤ 5.933834
s(29) ≥ 5.810000

30 =
s(30) ≤ 6
s(30) ≥ 5.883500

31 =
s(31) ≤ 6
s(31) ≥ 5.952500

32 O =
s(32) = 6

33 O =
s(33) = 6

34 O =
s(34) = 6

35 O =
s(35) = 6

36 O = R
s(36) = 6

37 =
s(37) ≤ 6.598620 deg 8
s(37) ≥ 6.440000

38 =
s(38) ≤ 6.707107 deg 2
s(38) ≥ 6.545000

39 =
s(39) ≤ 6.810723 deg 5
s(39) ≥ 6.650000

40 = R
s(40) ≤ 6.828428 deg 2
s(40) ≥ 6.700000

41 =
s(41) ≤ 6.926694 deg 42
s(41) ≥ 6.775000

42 =
s(42) ≤ 7
s(42) ≥ 6.847500

43 =
s(43) ≤ 7
s(43) ≥ 6.907500

44 =
s(44) ≤ 7
s(44) ≥ 6.972500

45 O =
s(45) = 7

46 O =
s(46) = 7

47 O =
s(47) = 7

48 O =
s(48) = 7

49 O = R
s(49) = 7

50 =
s(50) ≤ 7.571429
s(50) ≥ 7.400000

51 =
s(51) ≤ 7.690356 deg 2
s(51) ≥ 7.470000

52 =
s(52) ≤ 7.707107 deg 2
s(52) ≥ 7.550000

53 =
s(53) ≤ 7.822876 deg 2
s(53) ≥ 7.627500

54 =
s(54) ≤ 7.846668 deg 4
s(54) ≥ 7.685000

55 ≈
s(55) ≤ 7.945772
s(55) ≥ 7.728000

56 =
s(56) ≤ 8
s(56) ≥ 7.802500

57 =
s(57) ≤ 8
s(57) ≥ 7.872500

58 =
s(58) ≤ 8
s(58) ≥ 7.935000

59 O =
s(59) = 8

60 O =
s(60) = 8

61 O =
s(61) = 8

62 O =
s(62) = 8

63 O =
s(63) = 8

64 O = R
s(64) = 8

65 =
s(65) ≤ 8.535534 deg 2
s(65) ≥ 8.350000

66 =
s(66) ≤ 8.656855 deg 2
s(66) ≥ 8.430000

67 =
s(67) ≤ 8.707107 deg 2
s(67) ≥ 8.480000

68 =
s(68) ≤ 8.798796
s(68) ≥ 8.510000

69 =
s(69) ≤ 8.827195 deg 38
s(69) ≥ 8.620000

70 =
s(70) ≤ 8.880961
s(70) ≥ 8.657500

71 ≈
s(71) ≤ 8.944072
s(71) ≥ 8.721000

72 =
s(72) ≤ 9
s(72) ≥ 8.760000

73 =
s(73) ≤ 9
s(73) ≥ 8.813000

74 =
s(74) ≤ 9
s(74) ≥ 8.867500

75 =
s(75) ≤ 9
s(75) ≥ 8.940000

76 =
s(76) ≤ 9
s(76) ≥ 8.965000

77 O =
s(77) = 9

78 O =
s(78) = 9

79 O =
s(79) = 9

80 O =
s(80) = 9

81 O = R
s(81) = 9

82 =
s(82) ≤ 9.535534 deg 2
s(82) ≥ 9.320000

83 =
s(83) ≤ 9.634758deg 672
s(83) ≥ 9.370000

84 =
s(84) ≤ 9.698053
s(84) ≥ 9.411000

85 =
s(85) ≤ 9.742641 deg 2
s(85) ≥ 9.460000

86 =
s(86) ≤ 9.820566
s(86) ≥ 9.503000

87 =
s(87) ≤ 9.838816 deg 41
s(87) ≥ 9.580000

88 =
s(88) ≤ 9.882452
s(88) ≥ 9.620000

89 =
s(89) ≤ 9.949748 deg 2
s(89) ≥ 9.650000

90 =
s(90) ≤ 10
s(90) ≥ 9.730000

91 =
s(91) ≤ 10
s(91) ≥ 9.762500

92 =
s(92) ≤ 10
s(92) ≥ 9.770000

93 =
s(93) ≤ 10
s(93) ≥ 9.880000

94 =
s(94) ≤ 10
s(94) ≥ 9.950000

95 =
s(95) ≤ 10
s(95) ≥ 9.965000

96 =
s(96) ≤ 10
s(96) ≥ 9.970000

97 O =
s(97) = 10

98 O =
s(98) = 10

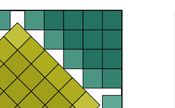
99 O =
s(99) = 10

100 O = R
s(100) = 10

The square packing problem asks for the side *s*(*n*) of the smallest square that can hold *n* unit squares, where the squares are free to rotate but cannot overlap

O proved optimal (45 of 100)
= exact value known (97 of 100)
≈ only known numerically (3 of 100)
R rigid (14 of 100)

recent result, since August, 2026 (81 of 100)



colors indicate distinct tilt angles
shade indicates number of full-side contacts
deg is the algebraic degree of that side length

Best packings due to Göbel, Trump, Wainwright, Hämäläinen, Stenlund, Friedman, Bidwell, Cantrell, Morandi, Ellsworth, Hajba, Schadt, Chaoweeraprasit, Couzo, Daniel, de Winter, Rehwaldt, Gupta, ry-xu, Chang, DeVincentis, hmbelvedere

Lower bounds due to Göbel, Stromquist, Kearney, Shiu, Bentz, Daniel, Vlasenko, Guzhou0806, Hosono, Ahmed, Karakuş, Ryu

Optimality proofs due to Göbel, Stromquist, Friedman, El Moumni, Kearney, Shiu, Bentz, Vlasenko, Levy, Daniel, Hosono, Ahmed, Karakuş; formalization contributions by Gurevitch, Guzhou0806; verification infrastructure and execution by EvolvingPrograms, Julian-JJ, Lewis

Diagram by Joshua Levy
October 10, 2026 · v0.5.0-c8abd5